

COM S 5310 Theory of Computing, Homework 4

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Due: November 10th, 11:59pm on Gradescope.

Problem 1.

Prove that $\Sigma_1^0 \subseteq \Sigma_2^0$

Answer.

Given the following rules:

$$\Sigma_1^0 = CE$$

$$\Pi_n^0 = co\Sigma_n^0$$

$$\Delta_n^0 = \Pi_n^0 \cap \Sigma_n^0$$

$$\Sigma_{n+1}^0 = \exists \Pi_n^0$$

Knowing that $\Sigma_1^0 = CE$, and

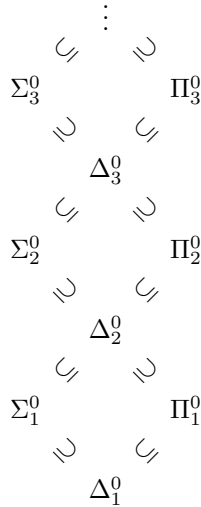
$$\Sigma_2^0 = \exists \Pi_1^0 = \exists (co\Sigma_1^0) = \exists coCE$$

We have to show $CE \subseteq \exists coCE$, which follows by:

$$\begin{aligned} DEC &\subseteq coCE \\ \implies \exists DEC &\subseteq \exists coCE \\ \implies CE &\subseteq \exists coCE \end{aligned}$$

Problem 2.

Prove all the inclusions in the infinite diagram.



Answer.

We will prove this question using induction. Specifically, we'll be showing the following inclusions:

- (1) $\Sigma_n^0 \subseteq \Pi_{n+1}^0$
- (2) $\Sigma_n^0 \subseteq \Sigma_{n+1}^0$
- (3) $\Sigma_n^0 \subseteq \Delta_{n+1}^0$
- (4) $\Pi_n^0 \subseteq \Pi_{n+1}^0$
- (5) $\Pi_n^0 \subseteq \Sigma_{n+1}^0$
- (6) $\Pi_n^0 \subseteq \Delta_{n+1}^0$
- (7) $\Delta_n^0 \subseteq \Delta_{n+1}^0$

Basis: $n = 1$

We have to prove the following:

- (1) $\Sigma_1^0 \subseteq \Pi_2^0$: based on lemma 25, If $\mathcal{C}$ is a class of languages that is closed under $\leq_m$ -reductions. Then

$$\mathcal{C} \subseteq \forall \mathcal{C} \cap \exists \mathcal{C}$$

using this lemma, since $\Sigma_1^0 = CE$ is closed under $\leq_m$ -reductions, $CE \subseteq \forall CE$. Thus,

$$\begin{aligned} CE \subseteq \forall CE &\iff CE \subseteq \text{coco}\forall CE \\ &\iff CE \subseteq \text{co}\exists \text{co}CE \\ &\iff \Sigma_1^0 \subseteq \text{co}\Sigma_2^0 \quad (\text{based on problem 1 } \Sigma_2^0 = \exists \text{co}CE) \\ &\iff \Sigma_1^0 \subseteq \Pi_2^0 \quad (\text{based on the definition } \Pi_2^0 = \text{co}\Sigma_2^0) \end{aligned}$$

- (2) $\Sigma_1^0 \subseteq \Sigma_2^0$: we have already proved this in the first part.
- (3) $\Sigma_1^0 \subseteq \Delta_2^0$: since $\Sigma_1^0 \subseteq \Sigma_2^0$ (based on 2) and $\Sigma_1^0 \subseteq \Pi_2^0$ (based on 1), thus, $\Sigma_1^0 \subseteq \Sigma_2^0 \cap \Pi_2^0 = \Delta_2^0$ (based on the definition).
- (4) $\Pi_1^0 \subseteq \Pi_2^0$: knowing that $CE = \Sigma_1^0 \subseteq \Sigma_2^0 = \exists \text{co}CE$ (based on problem 1), then

$$\begin{aligned} \Pi_1^0 \subseteq \Pi_2^0 &\iff \text{co}\Sigma_1^0 \subseteq \text{co}\Sigma_2^0 \\ &\iff \text{coco}\Sigma_1^0 \subseteq \text{coco}\Sigma_2^0 \\ &\iff \Sigma_1^0 \subseteq \Sigma_2^0 \end{aligned}$$

- (5) $\Pi_1^0 \subseteq \Sigma_2^0$: knowing that $\Sigma_1^0 \subseteq \Pi_2^0$ (based on 1), we have

$$\begin{aligned} \Sigma_1^0 \subseteq \Pi_2^0 &\implies \text{co}\Sigma_1^0 \subseteq \text{co}\Pi_2^0 \\ &\implies \Pi_1^0 \subseteq \text{coco}\Sigma_2^0 \\ &\implies \Pi_1^0 \subseteq \Sigma_2^0 \end{aligned}$$

- (6) $\Pi_1^0 \subseteq \Delta_2^0$: Since $\Pi_1^0 \subseteq \Sigma_2^0$ (based on 5) and $\Pi_1^0 \subseteq \Pi_2^0$ (based on 4), thus, $\Pi_1^0 \subseteq \Sigma_2^0 \cap \Pi_2^0 = \Delta_2^0$ (based on the definition).
- (7) $\Delta_1^0 \subseteq \Delta_2^0$: we know $\Delta_1^0 \subseteq \Pi_1^0$ (based on the definition), and $\Pi_1^0 \subseteq \Delta_2^0$ (based on 6), thus $\Delta_1^0 \subseteq \Delta_2^0$.

Induction Hypothesis: suppose the following properties hold for some $n \in \mathbb{N}$

- (1) $\Sigma_n^0 \subseteq \Pi_{n+1}^0$
- (2) $\Sigma_n^0 \subseteq \Sigma_{n+1}^0$
- (3) $\Sigma_n^0 \subseteq \Delta_{n+1}^0$

- (4) $\Pi_n^0 \subseteq \Pi_{n+1}^0$
- (5) $\Pi_n^0 \subseteq \Sigma_{n+1}^0$
- (6) $\Pi_n^0 \subseteq \Delta_{n+1}^0$
- (7) $\Delta_n^0 \subseteq \Delta_{n+1}^0$

Inductive Step $n \rightarrow n+1$: we will prove the following sequentially, using the results of parts 1, ..., $m-1$ when proving part m .

- (1) $\Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0$: since Σ_{n+1}^0 is closed under $\leq_m$ -reductions. Then

$$\Sigma_{n+1}^0 \subseteq \forall \Sigma_{n+1}^0$$

$$\begin{aligned} \Sigma_{n+1}^0 \subseteq \forall \Sigma_{n+1}^0 &\iff \Sigma_{n+1}^0 \subseteq \text{coco} \forall \Sigma_{n+1}^0 \\ &\iff \Sigma_{n+1}^0 \subseteq \text{co} \exists \text{co} \Sigma_{n+1}^0 \\ &\iff \Sigma_{n+1}^0 \subseteq \text{co} \exists \Pi_{n+1}^0 \quad (\text{based on the definition } \Pi_n^0 = \text{co} \Sigma_n^0) \\ &\iff \Sigma_{n+1}^0 \subseteq \text{co} \Sigma_{n+2}^0 \quad (\text{based on the definition } \exists \Pi_n^0 = \Sigma_{n+1}^0) \\ &\iff \Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0 \quad (\text{based on the definition } \Pi_{n+2}^0 = \text{co} \Sigma_{n+2}^0) \end{aligned}$$

Thus, $\Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0$.

- (2) $\Sigma_{n+1}^0 \subseteq \Sigma_{n+2}^0$:

$$\Sigma_{n+1}^0 \subseteq \Sigma_{n+2}^0 \iff \exists \Pi_n^0 \subseteq \exists \Pi_{n+1}^0$$

Since $\Pi_n^0 \subseteq \Pi_{n+1}^0$ based on the induction hypothesis, so is $\exists \Pi_n^0 \subseteq \exists \Pi_{n+1}^0$, and thus $\Sigma_{n+1}^0 \subseteq \Sigma_{n+2}^0$.

- (3) $\Sigma_{n+1}^0 \subseteq \Delta_{n+2}^0$:

Since $\Sigma_{n+1}^0 \subseteq \Sigma_{n+2}^0$ and $\Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0$, thus, $\Sigma_{n+1}^0 \subseteq \Sigma_{n+2}^0 \cap \Pi_{n+2}^0 = \Delta_{n+2}^0$.

- (4) $\Pi_{n+1}^0 \subseteq \Pi_{n+2}^0$:

$$\begin{aligned} \Pi_{n+1}^0 \subseteq \Pi_{n+2}^0 &\iff \text{co} \Sigma_{n+1}^0 \subseteq \text{co} \Sigma_{n+2}^0 \\ &\iff \text{co}(\exists \Pi_n^0) \subseteq \text{co}(\exists \Pi_{n+1}^0) \end{aligned}$$

Since $\Pi_n^0 \subseteq \Pi_{n+1}^0$ based on the induction hypothesis, so is $\exists \Pi_n^0 \subseteq \exists \Pi_{n+1}^0$ and $\text{co} \exists \Pi_n^0 \subseteq \text{co} \exists \Pi_{n+1}^0$, and thus $\Pi_{n+1}^0 \subseteq \Pi_{n+2}^0$.

- (5) $\Pi_{n+1}^0 \subseteq \Sigma_{n+2}^0$

knowing that $\Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0$, we have

$$\begin{aligned} \Sigma_{n+1}^0 \subseteq \Pi_{n+2}^0 &\implies \text{co} \Sigma_{n+1}^0 \subseteq \text{co} \Pi_{n+2}^0 \\ &\implies \Pi_{n+1}^0 \subseteq \text{coco} \Sigma_{n+2}^0 \\ &\implies \Pi_{n+1}^0 \subseteq \Sigma_{n+2}^0 \end{aligned}$$

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- (6) $\Pi_{n+1}^0 \subseteq \Delta_{n+2}^0$:
Since $\Pi_{n+1}^0 \subseteq \Sigma_{n+2}^0$ and $\Pi_{n+1}^0 \subseteq \Pi_{n+2}^0$, thus, $\Pi_{n+1}^0 \subseteq \Sigma_{n+2}^0 \cap \Pi_{n+2}^0 = \Delta_{n+2}^0$.
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- (7) $\Delta_{n+1}^0 \subseteq \Delta_{n+2}^0$, based on the definition, we know $\Delta_{n+1}^0 = \Pi_{n+1}^0 \cap \Sigma_{n+1}^0$, and thus

$$\Delta_{n+1}^0 \subseteq \Pi_{n+1}^0 \subseteq \Delta_{n+2}^0 \quad (\text{based on 6})$$

hence

$$\Delta_{n+1}^0 \subseteq \Delta_{n+2}^0.$$
